

Student Takeaway for Sync Session

<<Week 9: Applications of AI and ML - NLP>>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Applications of AI and ML - NLP		
Session duration	1 hour		
Session type	Lectures: Conceptual understanding of NLP tasks like text classification, translation, and sentiment analysis. Case Studies: Real-world applications of NLP in tools like WhatsApp, Google Search, YouTube, and chatbots.		
Scope	This session explores practical applications of NLP in daily life. It covers: • Key NLP tasks and techniques. • Tools and models powering NLP applications. • Case studies of NLP in popular digital tools. • Demonstrations of NLP capabilities and limitations.		
Learning objectives			
	Objective	Core capability	
	Understand NLP tasks and techniques	Real-world awareness of NLP applications	
	Apply NLP tools for text analysis	Ability to use NLP models for classification, translation, etc.	
	Explore NLP limitations and challenges	Practical understanding of AI's struggles with context, sarcasm, etc.	
	Connect NLP tools with everyday use cases	Problem-solving in language-based contexts	
Software/Tools	• Libraries: NLTK, spaCy, Hugging Face Transformers • Models: ChatGPT, Google Translate, Named Entity Recognition (NER) models • Datasets: Sample text for sentiment analysis, translation, summarisation • Presentation Tool: PowerPoint		

Pro tips for success:

Ask Bold Questions: Clarify how NLP handles ambiguity (e.g., "Apple" as fruit vs. company).

Be Hands-On: Experiment with live demos (e.g., translate a sentence to Hindi).

Collaborate: Discuss ethical concerns (e.g., bias in translation) with peers.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-3	Today's Focus	Introduction to NLP in daily life.	AI in WhatsApp, Google Search.
4	Where Do You See Language AI?	Brainstorming session: Spot NLP in your apps.	Spam detection, autocorrect.
5	Can You Guess What AI's Doing?	Interactive activity: Identify NLP tasks in examples.	Spam vs. intent detection.
6	NLP: How AI Understands Language	Key tasks: classification, NER, translation.	Sentiment analysis, summarisation.
7	NLP in Everyday Use	Real-life demos (sentiment, NER).	Positive/negative review analysis.
8	Sentiment Analysis Demo	AI judges emotions in text.	"Amazing phone" vs. "Cold service."
9	NER Demo	Extract names/places from text.	"Sachin Tendulkar in Mumbai."
10	Text Summarisation Demo	Shorten news articles.	Apple product launch summary.
11	Machine Translation Demo	Translate phrases (English → Hindi/Spanish).	Idioms like "raining cats and dogs."
12	Question Answering Demo	AI answers from context.	"Capital of France?" → "Paris."
13	AI as a Black Box	Input → Processing → Output workflow.	Bias in training data.
14	How to Imagine NLP Tasks	Analogies (e.g., "NER = highlighting text").	Sentiment analysis = thumbs up/down.
15	Funny and Tricky Mistakes	AI errors (sarcasm, idioms).	"Oh great, another rainy day!" → Positive.
16	Limitations of AI in NLP	Challenges: context, bias, cultural nuances.	Misinterpreted sarcasm examples.

Slide(s)	Topic/Issue	What to expect	Notes
17-18	From Rules to Learning Patterns	NLP evolution: rule-based → deep learning.	ChatGPT vs. early chatbots.
19	What Did We Learn Today?	Recap: Tasks, limitations, real-world use.	NLP in healthcare vs. entertainment.
20	Questions and Curiosities	Group reflection: Ethics, future trends.	"Will AI ever master sarcasm?"
21	Next Week: AI Beyond Text	Preview of computer vision/speech tech.	Facial recognition,
22	Q&A	Open discussion on NLP applications	Focus on real-world tools like chatbots, translation, and sentiment analysis.

Post session

Reflection challenge	• Question: Which NLP application (e.g., spam detection, translation, sentiment analysis) do you think has the biggest impact on daily life? Why? • Action: Jot down one idea for how you could use an NLP tool to solve a problem in your field.
Explore more	• Read: Articles about GPT-4's language capabilities or NLP in healthcare. • Watch: Tutorials on Hugging Face's sentiment analysis or building a chatbot with spaCy. • Experiment: Use Google Translate to convert a paragraph into 3 languages. Compare accuracy.
Get inspired	Did You Know? NLP powers real-time captioning for the hearing impaired and language learning apps like Duolingo. Your understanding of these tools can make technology more inclusive!
The journey ahead	Next Week: Dive into AI Beyond Text—explore how computers "see" (image recognition) and "hear" (speech-to-text).